

Battery

Installation

Safety

Batteries emit hydrogen gas which is highly inflammable and dangerous in closed spaces. Ensure the site is well ventilated.

To prevent current leakage to the ground, provide proper earth fault and insulate the battery stand.

www.relicellbattery.com



VRLA and Tubular batteries

VRLA Batteries - 12V 7AH to 250 AH
Tubular Batteries - 12V 40AH to 210 AH

Part of a Rs.1000 crore IT industry group
Tested and certified by Central Power
Research Institute, Bangalore (CPRI) and
Electronic Test and Development Centre
(STQC Directorate, Ministry of Information
Technology, Govt. of India). **Download Test
Certificates at www.relicellbattery.com**
UL Certified
VRLA Batteries approved and registered in
DGS&D for government purchases
47 Service Centres across India

RELICELL

Greenvision Technologies Pvt.Ltd

45/1, 4th Cross, Mallewaram, Bangalore - 560 003.
Tel: +91 080 4245 9459, Fax: +91 080 4245 9460.
Email: info@relicellbattery.com

Atmospheric Pressure Sensor

I am doing a project on weather monitoring system. But I could not find a suitable sensor and the corresponding driver circuit for atmospheric pressure measurement. Please tell me where can I find the atmospheric pressure sensor.

Abhisek Sahoo
Bhubaneswar

EFY: You may refer to 'Electronics For You Annual Guide 2010.' This directory contains contact details of manufacturers/distributors (in India) of various electronic equipment and devices including barometers, pressure sensors and manometers. You can purchase the directory from EFY associates Kits'n'Spares (address: D-88/5, Okhla Industrial Area, Phase 1, New Delhi 110020; Phone: (011) 26371661; E-mail: info@kitsnspares.com). The directory is also available online at 'www.eleb2b.com.'

Simulation Software

I love fiddling around with basic components and the oscilloscope. So I am looking for a good simulation program that could fulfil my needs. I have tried OrCAD Lite software but it lacks in some aspects.

I hope you will find a solution for me. I should be able to experiment with basic components such as resistors, capacitors, inductors and BJTs and also see the waveforms with the oscilloscope kind of functionality.

Vabhravahan Singh
Through e-mail

☐ I am an engineering student living in a remote place where components are unavailable. I want to design circuits of my own. For that, I need a software to draw circuits, verify the circuit outputs and display them in the form of waveform. Please help.

Saijanani Murali
Through e-mail

EFY: You may refer to VisualSpice, VisSim and EDWinx simulation software included in the CD supplied with EFY's Jan. 2009, May 2009 and Nov. 2009 issues, respectively. To purchase the CDs, contact Kits'n'Spares on 011-26371661 or e-mail to info@kitsnspares.com.

Robotics

I want to undertake a project on manual robot. Please suggest me some robotics projects published in EFY.

Shyam Sunder Gupta
Kolkata

EFY: You may refer to the following robotic projects published in EFY:

1. 'Automated Line-Following Robot' in Sep. 2009
2. 'Robotic Vision Using Matlab' in May 2009
3. 'PC-Based Mobile Robot for Navigation' in Oct. 2008
4. 'Cellphone-Operated Land Rover' in July 2008
5. 'Microcontroller-Based Infrared Tracking Robot' in April 2007
6. 'Remote-Controlled Land Rover' in June 2006

To purchase these issues, please contact Kits'n'Spares.

Developing PCBs

The article 'Developing PCBs Using Express PCB' published in February issue is truly a great job by Somnath Chakraborty. But I would like to add some information about the art of transferring conductor tracks to the paper and etchant solution. Use of a 130GSM (grams per square metre) photo paper is recommended, as it doesn't stick to the drum of the laser printer. Use of a transparent etchant solution, which consists of about 35 per cent of hydrochloric (HCL) acid and 15 per cent of hydrogen peroxide (H_2O_2), is also recommended. HCL acid is available in hardware stores by the name muriatic acid and H_2O_2 solution can be found in any pharmaceutical store. With this solution, it is easy to check the board during the etching process. The toner can be removed using petrol, paint remover or thinner.

Ronie Adhiraj Ghosh
Through e-mail

The author, Somnath Chakraborty, replies:

Actually, there are a number of ways to etch the PCB. The technique I have given is only one way to do it. Your suggestion is absolutely correct and useful. As regards the paper, 130GSM may not be easily available. So you can use a thinner 80GSM paper (EFY is printed on 60GSM paper) as mentioned in my article. It will also work.